Electrical Power and Energy Systems 120 (2020) 105990

Contents lists available at ScienceDirect

Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Fast grid synchronization method for variable-frequency AC systems

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ARTICLE INFO

Keywords:
Variable frequency
Positive-sequence voltage component
Negative-sequence voltage component
Fast dynamic response
More electric aircraft

ABSTRACT

Different from utility power systems, the grids of more electric aircraft (MEA) are frequency-variable, and the frequency change range is from 360 *Hz* to 800 *Hz*. Most grid synchronization methods are designed for constant-frequency grids whose frequency varies in a small range around 50 *Hz* or 60 *Hz*. Therefore, the dynamic performance of these methods decreases considerably in MEA grids. In addition, the frequency in MEA is higher than that in utility power systems. Grid voltage parameters are thus required to exhibit a fast dynamic response. To resolve this problem, this paper proposes an algebraic method to estimate the positive- and negative-sequence components (PNSC) of grid voltage accurately. By analyzing the relationship between grid voltage and its time delay signals, the linear regression model is established. Then, grid frequency, PNSC are obtained with an algebraic method. The proposed method relies on algebraic operations only and thus exhibits a fast dynamic response. Experiment results verify the effectiveness of the proposed method.

1. Introduction

Replacing conventional hydraulic and pneumatic systems by power electronic devices, more electric aircraft (MEA) achieve small volume and light weight. Hence, many increasing research interests are focused in this area [1]. The estimation of positive- and negative-sequence voltage components (PNSC) is indispensable to the power control and grid synchronization of MEA grids.

Utility power grids have a constant frequency of *50* or *60 Hz*. By contrast, MEA grids have a wide frequency range of *360 Hz* to *800 Hz*. Therefore, estimating PNSC in MEA grids is more difficult than that in utility power grids [2].

Numerous grid voltage parameter estimation methods have been proposed for constant-frequency AC systems. The main methods are Kalman filtering [3,4], discrete Fourier transform [5,6], zero-crossing detection [7], adaptive notch filter [8–13], least-squares estimation [14], and phase-locked loops (PLLs) [15–23]. Estimation parameters usually include PNSC, grid frequency, amplitudes, and phase angles. Among these estimation methods, PLL is the most frequently used approach to estimate grid voltage parameters of utility power grids due to its simple structure and good performance [24]. However, in applications with widely variable frequencies, such as MEA grids, the dynamic response of PLL in grid voltage parameter estimation decreases considerably.

For example, a synchronous reference frame PLL (SRF-PLL) was provided in [25] to estimate the voltage parameters of a three-phase constant-frequency AC grid with a small amount of calculation and a good parameter estimation effect. As a classical method in grid synchronization, SRF-PLL is frequently employed for the parameter estimation of utility power grids. But the stability and dynamic performance of SRF-PLL decrease remarkably when the frequency of AC grids varies dramatically. When AC grids are in unbalanced conditions, steady-state estimation accuracy also decreases. To improve the steady-state estimation accuracy of SRF-PLL under unbalanced conditions, [20] and [26] investigated an advanced SRF-PLL method and a decoupled double-synchronous reference frame phase-locked loop (DDSRF-PLL) for the extraction of PNSC via two synchronous reference frames rotating at positive- and negative-synchronous speeds. To improve the dynamic performance of SRF-PLL, [27] added a MAF (moving average filter) block to perform fast grid voltage parameter estimation. Furthermore, in [19,24], and [28–32], second-order generalized integrator PLL (SOGI-PLL), D-filter PLL, enhanced-PLL, quasi-type-1 PLL, and quadrature PLL were utilized to estimate the grid voltage parameters of constant-frequency AC systems and improve the steady-state performance and dynamic response. However, PLL-based methods only exhibit good estimation performance around the equilibrium point [24]. [24] provides the relationship between the dynamics and the frequency varying region of the conventional PLL. It concludes that PLL

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https://doi.org/10.1016/j.ijepes.2020.105990
Received 8 April 2019; Received in revised form 2 January 2020; Accepted 5 March 2020
Available online 13 April 2020
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has good performance only around the nominal grid frequency. However, when PLL work out of its phase locked region, its dynamics is weaken significantly. Compared with PLL, the proposed method is an algebraic method and hence has global stability.

In addition, another well-known voltage parameter estimation is so-called quarter cycle delay method, such as [33] and [34], which has fast dynamic response and simple structure. However, its dynamic is larger than a quarter cycle of the grid voltage. Also, this method has not been reported for the application of MEA grids. Moreover, [35] provied an algebraic method to estimate grid frequency rapidly. However, it can not estimate PNSC. Some other methods are also proposed for varying frequency grids, such as [36] and [37]. In [36], an observer-based method is addressed to achieve aircraft grid synchronization with good dynamic response. However, the usage of the observer increases the computational burden. A DFT-based frequency estimation method for aircraft grids is provided in [37], but the large computational burden is its major drawback.

In this paper, an algebraic method is proposed to estimate the frequency and PNSC in MEA power grids. A linear regression model is established based on a measurable signal, namely, grid voltage, by applying two time delay blocks. Then, an algebraic method is developed for the fast estimation of the PNSC of grid voltage. The proposed method involves several algebraic operators only and thus exhibits a fast dynamic response.

Compared with aforementioned estimations, the proposed method is an algebraic algorithm which does not rely on the differentials and integrals. It involves only basic trigonometry and algebra. Thus, it has superior performance in terms of dynamic response and steady-state accuracy, especially in widely variable frequency MEA grids.

2. Estimation of PNSC for MEA

In the utility grids, the frequency and PNSC can be accurately extracted by the conventional PLLs. However, for MEA grids, the dynamics of the conventional PLLs are not fast enough. To deal with this problem, an algebraic method is developed in this paper to estimate the frequency and PNSC with fast dynamics, which is suitable for MEA grids.

In order to develop this fast estimation, an algebraic model of the MEA grids is essential. Once the algebraic model is obtained, the estimation of the frequency and PNSC can be achieved by an algebraic relationship (18)-(22). In the following, the algebraic model is established based on the relationship among the measured voltages and their delay signals.

In MEA power systems, grid voltage is expressed as

$$u_{abc} = u_{abc_p} + u_{abc_n} + u_{abc_0} \tag{1}$$

where u_{abc} is the three-phase grid voltage, u_{abc_p}, u_{abc_n} and u_{abc_0} are the positive-, negative and zero-sequence voltage components, respectively. u_{abc_p}, u_{abc_n} and u_{abc_0} are described as

$$\begin{aligned} u_{abc_p} &= U_p \begin{bmatrix} \cos(\omega t + \varphi_p) \\ \cos\left(\omega t - \frac{2\pi}{3} + \varphi_p\right) \\ \cos\left(\omega t + \frac{2\pi}{3} + \varphi_p\right) \end{bmatrix} \\ u_{abc_n} &= U_n \begin{bmatrix} \cos(\omega t + \varphi_n) \\ \cos\left(\omega t + \frac{2\pi}{3} + \varphi_n\right) \\ \cos\left(\omega t - \frac{2\pi}{3} + \varphi_n\right) \end{bmatrix} \\ u_{abc_0} &= U_0 \begin{bmatrix} \cos(\omega t + \varphi_0) \\ \cos(\omega t + \varphi_0) \\ \cos(\omega t + \varphi_0) \end{bmatrix} \end{aligned} \tag{2}$$

where U_p, U_n, U_0, φ_p, φ_n and φ_0 are the amplitudes and phase angles of the positive-, negative and zero-sequence voltage components. ω is the frequency. The zero-sequence voltage components can be obtained by $u_{abc_0} = [(u_a+u_b+u_c)/3,\ (u_a+u_b+u_c)/3,\ (u_a+u_b+u_c)/3]^{\mathrm{T}}$. In the following, this paper focuses on the estimation of PNSC.

Using

$$u_{\alpha\beta} = T_{\alpha\beta} u_{abc} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} u_{abc} \tag{3}$$

u_{abc} is transformed into

$$u_{\alpha\beta} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} u_{\alpha_p} + u_{\alpha_n} \\ u_{\beta_p} + u_{\beta_n} \end{bmatrix} \tag{4}$$

where

$$\mathrm{u}_{\alpha_\mathrm{p}} = \mathrm{U}_p \cos(\omega t + \varphi_p) = \mathrm{U}_p \cos\phi_p \tag{5}$$

$$\mathrm{u}_{\alpha_\mathrm{n}} = \mathrm{U}_n \cos(\omega t + \varphi_n) = \mathrm{U}_n \cos\phi_n \tag{6}$$

$$\mathrm{u}_{\beta_\mathrm{p}} = \mathrm{U}_p \sin(\omega t + \varphi_p) = \mathrm{U}_p \sin\phi_p \tag{7}$$

$$\mathrm{u}_{\beta_\mathrm{n}} = -\mathrm{U}_n \sin(\omega t + \varphi_n) = -\mathrm{U}_n \sin\phi_n \tag{8}$$

For convenience, it is defined that

$$u_x(t) = u_{x_p} + u_{x_n} = U_m \sin(\omega t + \varphi_m) \tag{9}$$

where the subscript x represents α and β.

After transferring delay blocks, it obtains

$$\begin{cases} v_{x1}(t) = u_x(t - T_s) = U_m \sin(\omega t - \omega T_s + \varphi_m) \\ v_{x2}(t) = u_x(t - 2T_s) = U_m \sin(\omega t - 2\omega T_s + \varphi_m) \end{cases} \tag{10}$$

where T_s is the sampling period.

Expanding Eqs. (9) and (10) results in the following:

$$\begin{cases} v_{x1}(t) = \gamma_1 U_m \sin(\omega t + \varphi_m) - \gamma_2 U_m \cos(\omega t + \varphi_m) \\ v_{x2}(t) = \gamma_3 U_m \sin(\omega t + \varphi_m) - \gamma_4 U_m \cos(\omega t + \varphi_m) \end{cases} \tag{11}$$

where

$$\begin{aligned} &\gamma_1 = \cos(\omega T_s) \quad \gamma_2 = \sin(\omega T_s) \\ &\gamma_3 = \cos(2\omega T_s) \quad \gamma_4 = \sin(2\omega T_s) \end{aligned} \tag{12}$$

The relationships among γ_1, γ_2, γ_3 and γ_4 are expressed as

$$\gamma_3 = 2\gamma_1^2 - 1 \quad \gamma_4 = 2\gamma_1\gamma_2 \tag{13}$$

Given that $0 < \omega T_s < \pi$, it obtains

$$\gamma_2 \neq 0 \tag{14}$$

Substituting Eq. (9) into Eq. (11) produces

$$\begin{cases} \gamma_2 U_m \cos(\omega t + \varphi_m) = \gamma_1 u_x(t) - v_{x1}(t) \\ \gamma_4 U_m \cos(\omega t + \varphi_m) = \gamma_3 u_x(t) - v_{x2}(t) \end{cases} \tag{15}$$

From Eqs. (13), (14), and (15), it is derived

$$u_x(t) + v_{x2}(t) = 2\gamma_1 v_{x1}(t) \tag{16}$$

$\hat{\gamma}_1$ satisfies

$$\hat{\gamma}_1 = \frac{u_x(t) + v_{x2}(t)}{2v_{x1}(t)} \quad \text{for any } v_{x1}(t) \neq 0 \tag{17}$$

By considering Eq. (12), the frequency of the MEA grid is obtained as follows:

$$\hat{\omega} = \frac{\arccos(\hat{\gamma}_1)}{T_s} \tag{18}$$

With Eq. (12), it is deduced

$$\hat{\gamma}_2 = \sin(\hat{\omega} T_s) = \sin(\arccos(\hat{\gamma}_1)) \tag{19}$$

The in-quadrature output signals of the proposed estimator in Fig. 1,

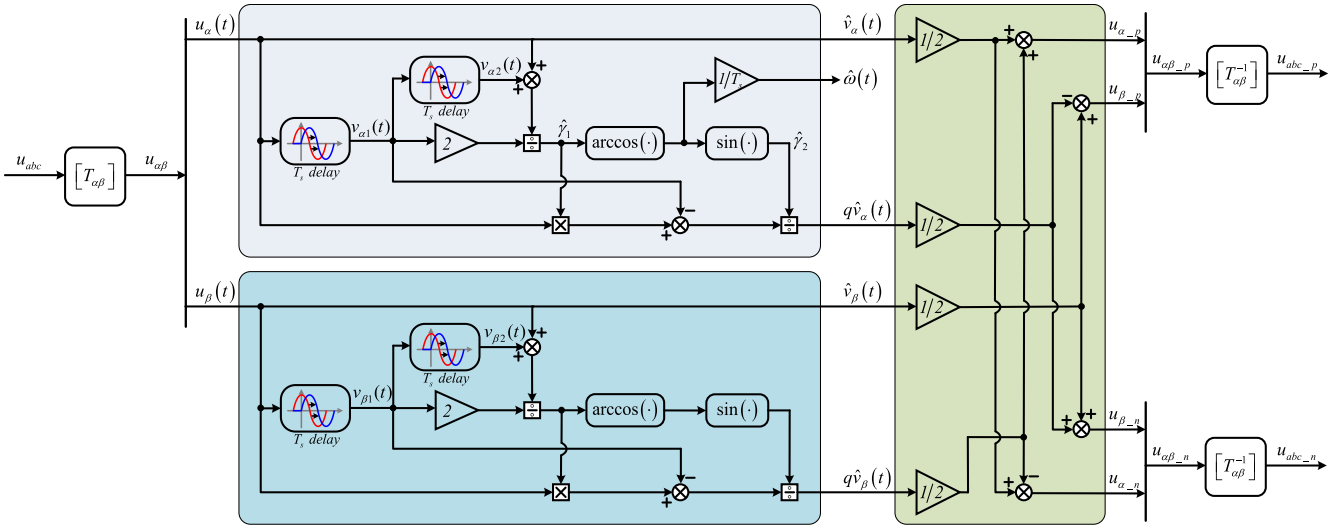


Fig. 1. Control diagram of the propsoed method.

that is, $\hat{v}_x(t)$ and $q\hat{v}_x(t)$, are defined by the following functions.

$$\begin{cases} \hat{v}_x(t) = u_x(t) = U_m \sin(\omega t + \varphi_m) \\ q\hat{v}_x(t) = U_m \cos(\omega t + \varphi_m) \end{cases} \tag{20}$$

Eqs. (11) and (19) result in

$$q\hat{v}_x(t) = U_m \cos(\omega t + \varphi_m) = \frac{\hat{\gamma}_1 u_x(t) - v_{x1}(t)}{\hat{\gamma}_2} \tag{21}$$

Then, the PNSC can be estimated by the relations

$$\begin{aligned} \hat{u}_{\alpha_p} &= \frac{\hat{v}_\alpha + q\hat{v}_\beta}{2},\ \hat{u}_{\alpha_n} = \frac{\hat{v}_\alpha - q\hat{v}_\beta}{2} \\ \hat{u}_{\beta_p} &= \frac{-q\hat{v}_\alpha + \hat{v}_\beta}{2},\ \hat{u}_{\beta_n} = \frac{q\hat{v}_\alpha + \hat{v}_\beta}{2} \end{aligned} \tag{22}$$

These signals in (22) are realized in the αβ-reference frame. If necessary, they can also be transformed into abc-reference frame.

Fig. 1 presents a control diagram of the proposed method. Three signals including $\hat{v}_x(t)$, $q\hat{v}_x(t)$, and $\hat{\omega}$, can be estimated with (18)-(22). They are calculated by signal $\hat{u}_x(t)$ and its delay. These calculations involve basic trigonometry and algebra. Thus, they are easy to implement on a standard digital hardware platform, such as DSP. Unlike the conventional PLLs, the proposed method (18)–(22) only contains algebric operators without differentials and integrals. Hence, it has fast dynamic responses and is more suitable for MEA grids.

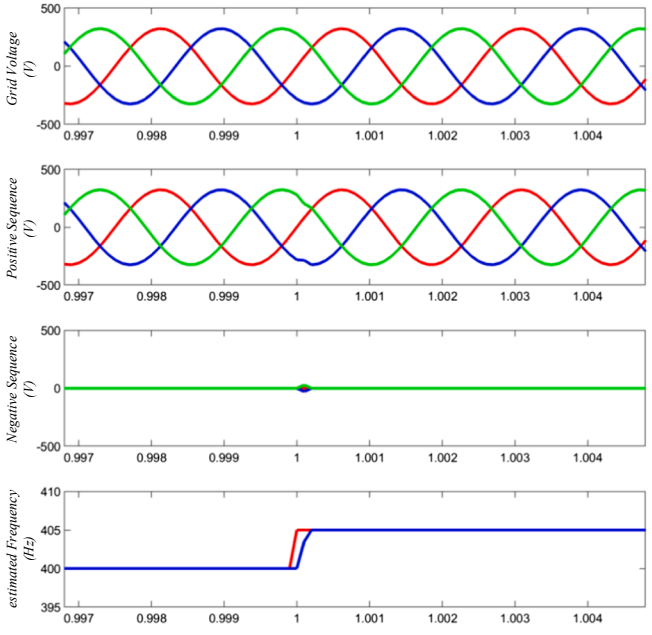


Fig. 2. Simulation Results of the proposed method: case of frequency jump from *400 Hz* to *405 Hz*.

3. Simulation results

Different from utility power systems, MEA grids are frequency-variable, and the frequency change range is from 360 *Hz* to 800 *Hz*. Higher frequency needs faster parameters estimation. In this paper, we focus on the fast estimation method of frequency, positive- and negative-sequence components of grid voltage. The proposed method is implemented in Matlab/Simulink to verify the effectiveness and the dymanic.

A simulation is tested under a small frequency jump from 400 *Hz* to 405 *Hz*, the simulation results is shown in Fig. 2. From Fig. 2, the

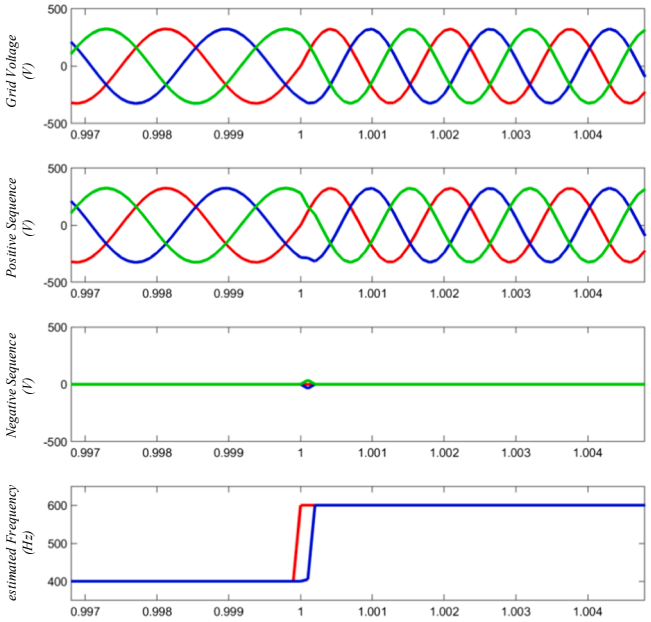


Fig. 3. Simulation results of the proposed method: case of frequency jump from *400 Hz* to *600 Hz*.

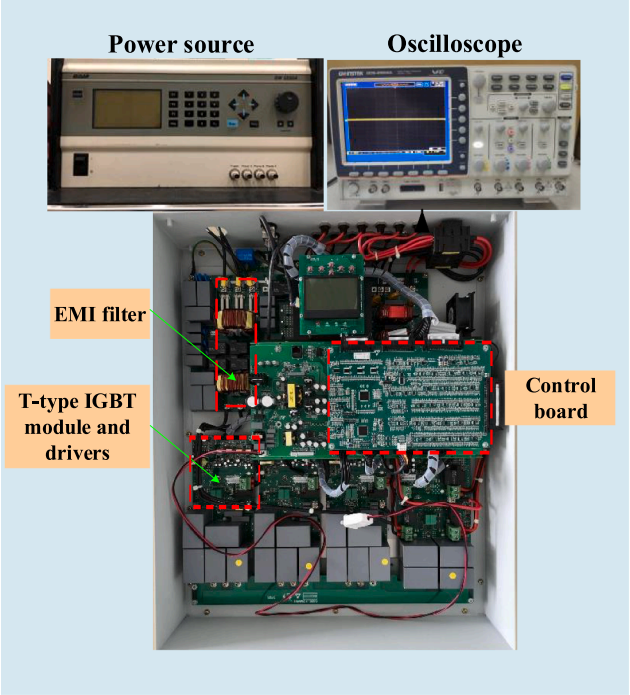


Fig. 4. Structure of the experimental setup.

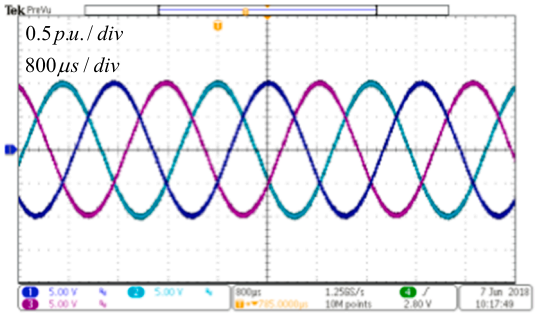


Fig. 5. Grid voltage: case of frequency jump from *400 Hz* to *405 Hz*.

proposed method estimates the PNSC with zero steady-state error. The 5-*Hz* frequency jump causes a small transient about 50 *us*.

Fig. 3 shows the simulation results when the frequency jumps from 400 *Hz* to 600 *Hz*. When t less than 1 *s*, the 400 *Hz* is accurately estimated. At t = 1 *s*, the 200-*Hz* frequency jump occurs, it results in a 50 *us*-dynamic. Then, the proposed method achieves the frequency estimation with zero steady-state error. In addition, the proposed method estimates PNSC accurately.

4. Experiment results

The experiment results are provided to verify the proposed method. SOGI-PLL and the proposed method are implemented in a control board with TMS320F28335 DSP, as shown in Fig. 4. The nominal parameters of MEA power systems are $\omega_r = 400 \times 2\pi\ rad/s$ and $U_r = 230\sqrt{2}\ V$ (*1.0 p.u.*). The advantages of the proposed method are highlighted by comparing with SOGI-PLL in four cases. The control parameters of SOGI-PLL are set as $K_P = 1104$ and $K_I = 16928$ [24].

4.1. A. Small frequency jump

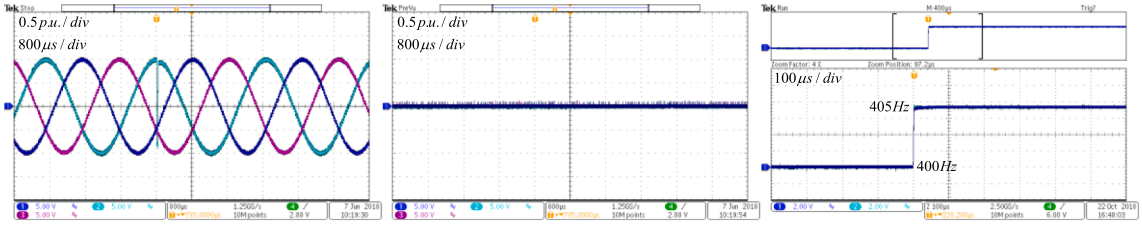
Fig. 5 shows the small frequency jump from *400 Hz* to *405 Hz*.

Figs. 6–7 illustrate the estimation of the PNSC and their frequency in this case. It is observed from Fig. 6 that the small frequency jump results in a slight transient which is less than 100 *us*. In addition, the proposed method estimate PNSC with zero steady-state error. When compared with SOGI-PLL shown in Figs. 6(c) and 7(c), the proposed method has faster dynamic response. In details, the proposed method estimates the grid voltage parameters without a steady-state error in a nearly *40* times shorter time. For clarity, the time scales and *Hz/div* ratios in Figs. 6(c) and 7(c) are different.

4.2. Large frequency jump

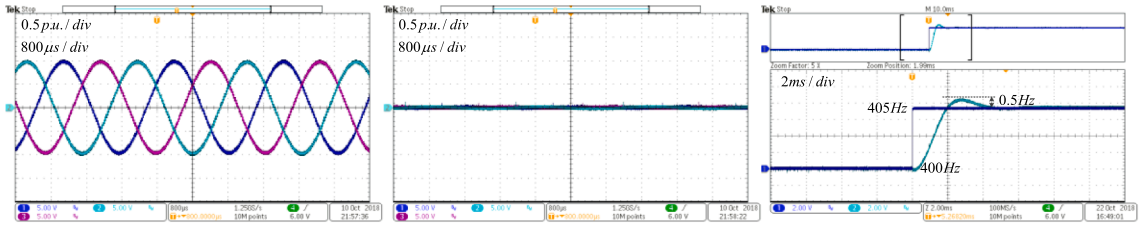
Similar to the situation in *Case A*, the grid voltage in *Case B* is formed by a *1.0 p.u.* positive-sequence voltage. The grid voltage has a large frequency jump from *400 Hz* to *600 Hz*. The grid voltage in case B is shown in Fig. 8.

The experiment results on grid voltage, PNSC, and frequency estimated by the proposed method are provided in Fig. 9. The proposed method estimates the PNSC accurately. Fig. 10 presents the SOGI-PLL results of the estimated PNSC and their frequencies. Compared the



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 6. Performance of the proposed method: case of frequency jump from *400 Hz* to *405 Hz*.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 7. Performance of SOGI-PLL: case of frequency jump from *400 Hz* to *405 Hz*.

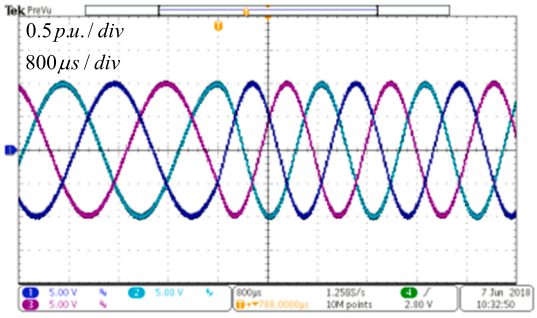


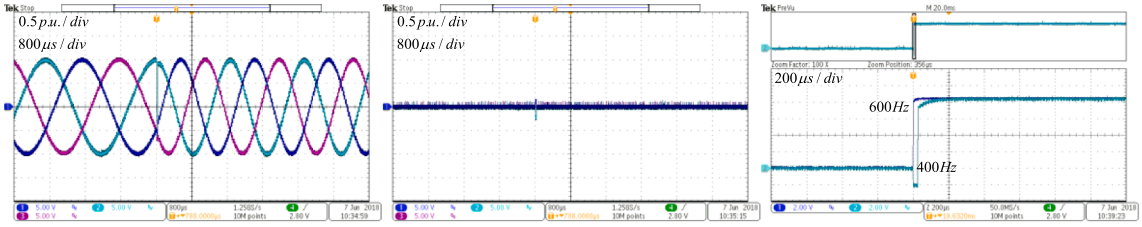
Fig. 8. Grid voltage: case of frequency jump from *400 Hz* to *600 Hz*.

Notably, the proposed method obtains a fast and precise estimation of grid frequency at this point, even under the amplitude jump condition. Such a fast and precise estimation cannot be obtained by the conventional SOGI-PLL.

4.4. Phase angle jump

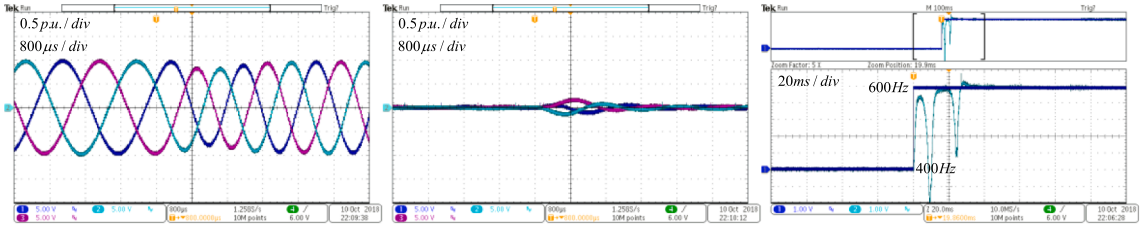
To confirm the benefits of the proposed method, we implement a $\pi/4$ *rad*-phase angle jump case in the experimental platform. Fig. 14 illustrates the results of grid voltage in this case.

Fig. 15(a) and (b) show that the proposed method estimates the PNSC with a zero steady-state error. As indicated in Fig. 15(c), the accuracy of frequency estimation is unaffected by the phase angle jump.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 9. Performance of the proposed method: case of frequency jump from *400 Hz* to *600 Hz*.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 10. Performance of SOGI-PLL: case of frequency jump from *400 Hz* to *600 Hz*.

experimental results in Figs. 9 and 10, it is further confirmed that the proposed method has better performance in dynamic response. When the frequency changes largely, the performance of SOGI-PLL deteriorates in terms of steady-state accuracy and dynamic response, while the proposed method rapidly estimates the MEA voltage parameters without a steady-state error.

4.3. Amplitude jump

In this case, the positive-sequence voltage has a 0.25 *p.u.* drop and a 0.25 *p.u.* negative-sequence voltage is added into the grid. Fig. 11 shows the grid voltage results under these amplitude jump conditions.

Fig. 12 shows the experiment results on the positive- and negative-sequence voltages and frequency estimated by the proposed method. Fig. 13 shows the experiment results on the PNSC and frequency estimated by SOGI-PLL.

The plots in Fig. 12(a) and (b) show that the estimation of PNSC is nearly perfect. On the contrary, Fig. 13(a) and (b) indicate that although the PNSC are estimated by SOGI-PLL without a steady-state error, the dynamic response during estimation is slow. Figs. 12(c) and 13(c) show the evolution of the grid frequency estimated by the proposed algorithm and SOGI-PLL, respectively.

Fig. 16 shows the experimental results obtained by SOGI-PLL. From the experimental results shown in Fig. 16, we derive the following observations.

1) When SOGI-PLL is used, the estimations of PNSC have a zero steady-state error.
2) Phase angle jump causes a transient (approximately *2.4 ms*) in the estimations of frequency.

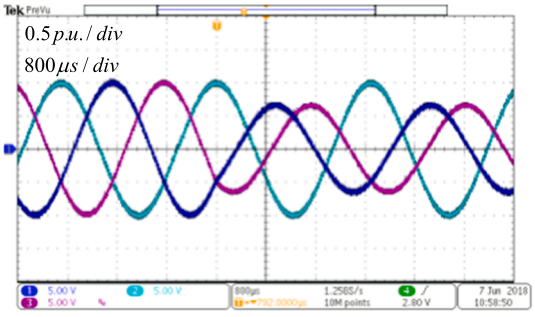
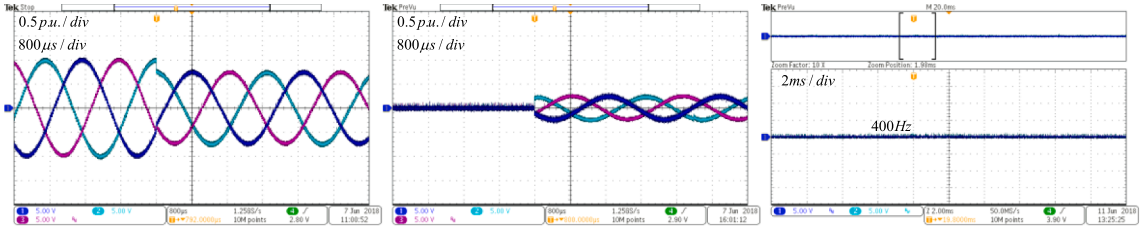
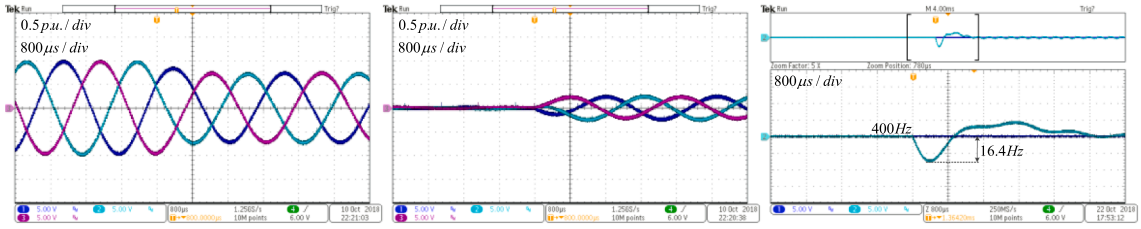


Fig. 11. Grid voltage: case of amplitude jump.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 12. Performance of the proposed method: case of amplitude jump.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 13. Performance of SOGI-PLL: case of amplitude jump.

The experimental results shown in Figs. 15 and 16 indicate that the proposed algorithm can achieve an more accurate estimation compared with SOGI-PLL.

Table 1 provides a dynamic performance summary of the proposed method and SOGI-PLL. The proposed method has a faster settling time and a smaller overshoot than SOGI-PLL.

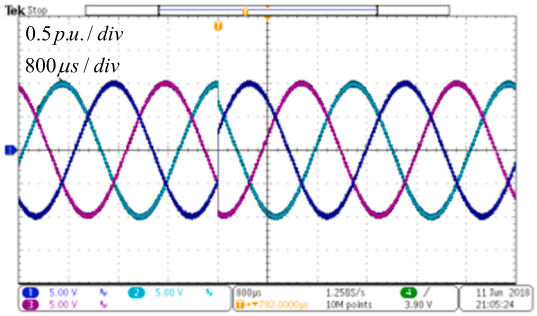
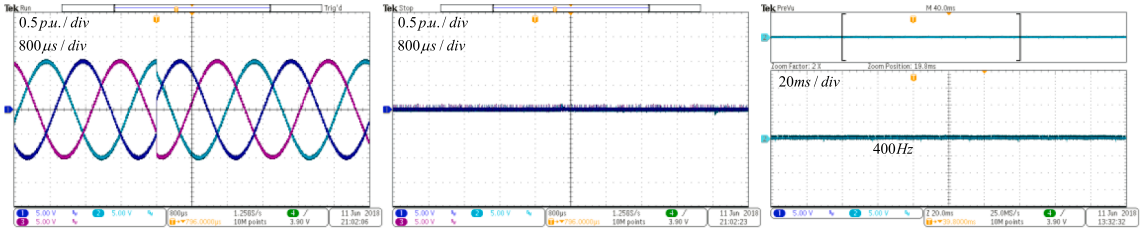


Fig. 14. Grid voltage: case of phase angle jump.

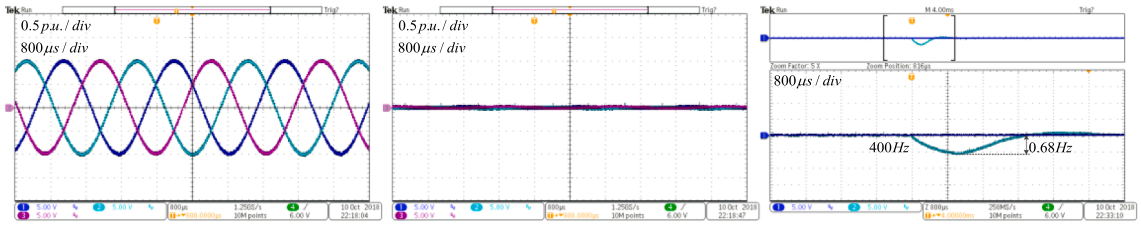
5. Conclusion

Different from most grid synchronization methods that are designed for constant-frequency power systems and have slow dynamic responses in MEA grids, the grid synchronization method proposed in this paper rapidly estimates the grid voltage parameters of MEA. A linear regression model is established based on the measurable grid voltage and its time delay signals. An algebraic method is developed to estimate the frequency, PNSC of grid voltage. Given that the proposed method only involves algebraic operators, it has a fast dynamic response. The experiment results reveal the method's superior performance in MEA grids.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 15. Performance of the proposed method: phase angle jump.



(a) positive-sequence voltage (b) negative-sequence voltage (c) estimated frequency

Fig. 16. Performance of SOGI-PLL: phase angle jump.

Table 1
Performance of estimated frequency under different cases.

Case	Settling time (*ms*)		Maximal overshoot (*Hz*)	
	Proposed method	SOGI-PLL	Proposed method	SOGI-PLL
Case A	0.08	4.2	0	0.5
Case B	0.08	36.0	–50	−84
Case C	0	4	0	−16.4
Case D	0	2.4	0	−0.68

CRediT authorship contribution statement

Mingdi Fan: Methodology, Supervision. **Zhiyong Dai:** Methodology, Supervision. **Yong Yang:** Validation. **Menxi Xie:** Validation. **Peiqi Zhao:** Software. **Juxiang Zhang:** Data Curation.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Fast Grid Synchronization Method for Variable-Frequency AC Systems.”

Acknowledgement

This work was supported in part by the National Natural Science Foundation of China under Grants 51907137, 51707140, 51977136 and 51707127, the Funds of Key Laboratory of Industrial Internet of Things and Networked Control under Grants 2018FF01, the Open Research Fund of National Rail Transportation Electrification and Automation Engineering Technology Research Center under Grants NEEC-2019-B07 and NEEC-2019-B08.

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